

Telecom Customer Churn Prediction

End-to-End Analytics Project Report

Data Cleaning · Statistical Analysis · Machine Learning · Business Recommendations

7,043

Dataset Records

87%

Model Accuracy

83%

Model Recall

~\$722K

Recoverable Revenue

1. Problem Statement, Goals & Success Criteria

Customer churn—the rate at which subscribers cancel or do not renew their service—is one of the most significant and costly challenges facing the telecommunications industry. Acquiring a new customer costs five to seven times more than retaining an existing one, making early identification of at-risk customers a high-priority business objective.

Business Problem

The telecom provider is experiencing an overall churn rate of approximately 26.5%, with disproportionately high attrition among month-to-month contract holders and fiber optic internet subscribers. Without a systematic, data-driven early-warning system, the retention team is reacting to churn rather than preventing it.

Project Goals

- Identify the structural and behavioural drivers of churn through rigorous exploratory data analysis.
- Build a predictive model capable of flagging high-risk customers before they cancel.
- Quantify the financial impact of churn and estimate the recoverable revenue from proactive retention.
- Deliver actionable, operationally viable recommendations to the retention and product teams.

Intended Audience

Executive Leadership	High-level revenue-at-risk KPIs, ROI of retention investment, strategic contract and pricing recommendations.
Retention / CX Team	Prioritised customer lists ranked by model-generated churn-risk scores; scripted intervention playbooks.

Product & Pricing	Service-tier churn rates (Fiber vs DSL), satisfaction gaps, and pricing sensitivity insights.
Data / Analytics	Model methodology, feature engineering decisions, pipeline architecture, and re-training schedule.

Criteria for Success

Model Recall Target	> 80% — catch the majority of true churners before they leave.
Model Accuracy Target	> 85% — minimise false positives that waste retention budget.
Revenue Recovery Target	Identify at minimum \$500K in recoverable annual revenue.
Operational Integration	Model outputs delivered as a ranked flag list usable in the CRM within 30 days.
Dashboard Adoption	Tableau suite refreshed weekly; used by retention team within 60 days of launch.

Approach Overview

The project follows a four-stage end-to-end pipeline: (1) SQL-based data cleaning and exploratory analysis to establish a trusted single source of truth; (2) statistical pattern analysis to surface key churn drivers; (3) competitive machine learning modelling using a Random Forest Classifier selected for its superior recall performance; and (4) deployment of findings into a Tableau dashboard suite and a set of ranked, actionable business recommendations.

2. Data Sources, Cleaning & Data Dictionary

Data Source

Source	IBM Telco Customer Churn Dataset (publicly available benchmark dataset).
Records	7,043 customer rows representing a snapshot of a single regional telecom provider.
Time Period	Cross-sectional snapshot; tenure spans 0–72 months.
Format	Single flat CSV file ingested into a SQL database for cleaning and EDA.
Target Variable	Churn (Yes / No) — positive class rate: 26.5% (1,869 churned customers).

Data Cleaning Summary

All cleaning steps were enforced in SQL prior to any modelling to ensure a single, auditable source of truth. No data was modified post-modelling.

NULL / Empty Correction	11 records contained empty strings in TotalCharges (all new customers with Tenure = 0). Converted to 0.00 via SQL CASE statement. No records dropped.
Duplicate Detection	CustomerID verified as unique across all 7,043 rows. No duplicate rows found.
Data Type Enforcement	TotalCharges and MonthlyCharges cast to FLOAT. SeniorCitizen cast from INTEGER (0/1) to VARCHAR (No/Yes) for consistency with other binary flags.
Outlier Review	MonthlyCharges and TotalCharges reviewed via IQR method. No extreme outliers requiring removal were identified; high charges are legitimate for long-tenure customers.
Categorical Standardisation	All string fields trimmed of leading/trailing whitespace. Category labels verified for consistency (e.g. 'No internet service' used consistently across dependent service columns).

Data Dictionary

Field	Type	Description	Notes
CustomerID	String	Unique customer identifier	No nulls; primary key

Gender	<i>String</i>	Male / Female	<i>Balanced distribution</i>
SeniorCitizen	<i>Integer</i>	0 = not senior, 1 = senior	<i>Binary flag</i>
Partner	<i>String</i>	Whether customer has a partner (Yes/No)	
Dependents	<i>String</i>	Whether customer has dependents (Yes/No)	
Tenure	<i>Integer</i>	Months as a customer (0 – 72)	<i>0 = new customers</i>
PhoneService	<i>String</i>	Phone line subscription (Yes/No)	
MultipleLines	<i>String</i>	Multiple phone lines (Yes/No/No phone service)	
InternetService	<i>String</i>	DSL / Fiber optic / No	<i>Key churn driver</i>
OnlineSecurity	<i>String</i>	Yes / No / No internet service	
TechSupport	<i>String</i>	Yes / No / No internet service	<i>Linked to churn</i>
Contract	<i>String</i>	Month-to-month / One year / Two year	<i>Strongest predictor</i>
PaperlessBilling	<i>String</i>	Yes / No	
PaymentMethod	<i>String</i>	Electronic / Mailed check / Auto bank / Auto CC	
MonthlyCharges	<i>Float</i>	Current monthly charge in USD	<i>Range \$18–\$119</i>
TotalCharges	<i>Float</i>	Total billed to date in USD	<i>Empty -> 0.00 (new)</i>
Churn	<i>String</i>	Target variable – Yes / No	<i>26.5% positive rate</i>

All 17 fields were retained after cleaning. No columns were dropped; however, CustomerID was excluded from modelling as a non-predictive identifier.

3. Patterns, Trends & Statistical Insights

Key Statistical Findings

Segment	Churn Rate	Risk Level	Recommended Action
Month-to-Month Contract	43%	Critical	Offer annual upgrade incentive
Fiber Optic Internet	~30%	High	Service quality review + discount
DSL Internet	~15%	Moderate	Loyalty rewards programme
One-Year Contract	11%	Low-Moderate	Nudge toward two-year renewal
Two-Year Contract	3%	Low	Maintain satisfaction touchpoints

Churn by Contract Type

Contract type is the single most structurally important churn predictor. Month-to-month customers churn at 43%—more than fourteen times the rate of two-year contract holders (3%). This is not primarily a satisfaction issue; it reflects the absence of commitment friction. Customers on flexible contracts have a low cost of switching and respond strongly to competitor promotions.

Churn by Internet Service Type

Fiber optic subscribers churn at approximately twice the rate of DSL users. This is counter-intuitive given that fiber is a premium service, but it likely reflects: (1) higher monthly charges creating stronger price sensitivity; (2) elevated service quality expectations that are not consistently met; and (3) greater competition in the fiber segment from alternative providers.

Churn by Tenure

The first 12 months of a customer's life are the highest-risk period. Customers who survive beyond 24 months exhibit sharply lower churn probability, suggesting that loyalty inertia builds significantly over time. Early onboarding quality and the first billing cycle are critical intervention points.

Churn by Monthly Charges

Customers paying above \$65/month churn at a materially higher rate than those below \$45/month. This charge-sensitivity effect is strongest among month-to-month customers, indicating that perceived value for money—not absolute price—is the driver. Customers who bundle multiple services churn less despite paying more overall.

Dashboard & Visualisation Suite

Findings were operationalised in a three-panel Tableau dashboard suite:

Executive Overview	High-level KPIs: total churn rate, revenue at risk, monthly trend line, and top-3 churn drivers. Designed for weekly senior leadership review.
--------------------	--

Geographic Mapping	Zip-code-level choropleth identifying clusters with above-average churn probability. Enables targeted regional retention campaigns.
Operational Flagging	Customer-level table ranked by model-generated risk score. Integrated with CRM to trigger daily outreach queues for the retention team. Refreshed weekly.

4. Machine Learning Pipeline & Predictive Model

Model Selection Rationale

Two classification algorithms were trained and compared. Recall was the primary optimisation metric because the business cost of a missed churner (lost lifetime value) significantly exceeds the cost of a false positive (an unnecessary retention call). The Random Forest Classifier was selected for production deployment.

Model	Accuracy	Recall	F1-Score	Notes
Logistic Regression	80%	62%	70%	Baseline – interpretable
Random Forest ✓	87%	83%	85%	Selected – best recall

The Random Forest model achieved 87% accuracy and 83% recall—a 21-point recall improvement over the logistic regression baseline. F1-Score of 85% indicates a strong balance between precision and recall.

Preprocessing & Feature Engineering

- Categorical encoding: all string features one-hot encoded; SeniorCitizen binary flag retained.
- Class imbalance handling: SMOTE (Synthetic Minority Oversampling) applied to training set to address the 73.5% / 26.5% class split.
- Train/test split: 80% training, 20% holdout test set; stratified by churn label.
- Hyperparameter tuning: 5-fold cross-validated GridSearch over n_estimators (100–500), max_depth (5–20), and min_samples_leaf (1–10).
- Optimal parameters: n_estimators = 300, max_depth = 15, min_samples_leaf = 2.

Feature Importance (Random Forest)

Tenure		21%
Monthly Charges		19%
Contract Type		17%
Internet Service		13%
Tech Support		10%
Payment Method		8%
Senior Citizen		7%
Paperless Billing		5%

Tenure (21%), Monthly Charges (19%), and Contract Type (17%) collectively account for 57% of total model importance—confirming the statistical findings in Section 3.

Predictive Application

The trained model is applied to the full customer base on a weekly refresh cycle. Each customer receives a churn probability score (0–100). Customers scoring above 70 are flagged as High Risk and surface in the Tableau Operational dashboard for immediate retention outreach. At current churn levels, approximately 580 customers per week meet this threshold.

5. Business Impact & Recommendations

Revenue at Risk

Total Churned Customers (Dataset)	1,869 customers (26.5% of base)
Average Monthly Charges	\$64.76 per customer
Estimated Annual Revenue Lost	~\$1.45M ($1,869 \times \64.76×12)
Model-Identifiable High-Risk Pool	~1,550 customers (83% recall applied)
Assumed Retention Success Rate	~30% with proactive intervention (industry benchmark)
Estimated Recoverable Revenue	~\$722K annually ($1,550 \times 30\% \times \64.76×12)

Strategic Recommendations

R1: Contract Migration Campaign

HIGH

Offer month-to-month customers a 10–15% discount to migrate to annual contracts. Target the ~2,200 month-to-month subscribers with tenure < 12 months first. Even a 20% conversion rate would remove ~440 customers from the highest-risk tier.

R2: Fiber Optic Value Programme

HIGH

Conduct NPS survey among fiber subscribers to identify specific dissatisfaction drivers. Bundle one free service add-on (e.g. Online Security or Tech Support) for fiber customers with tenure < 6 months to increase perceived value and reduce early churn.

R3: First-90-Day Onboarding Programme

HIGH

Create a structured 90-day onboarding journey for new customers with proactive check-in calls at Day 7, Day 30, and Day 90. New customers (Tenure = 0–3 months) represent the highest per-period churn risk.

R4: Model-Driven Weekly Outreach

MEDIUM

Operationalise the Random Forest risk scores via the Tableau Operational dashboard. Retention agents should work through the top-risk queue daily, prioritising customers with scores > 70 and monthly charges > \$65.

R5: Pricing Sensitivity Review

MEDIUM

Conduct an elasticity analysis to determine optimal price points for high-charge segments. Introduce a mid-tier bundle (DSL + Security + one add-on) priced at \$55–\$60 as an upgrade path for price-sensitive customers.

6. Setbacks, Limitations & Further Data Required

Known Limitations

- Cross-sectional data only: The dataset represents a single point-in-time snapshot. Without longitudinal transaction data, it is impossible to model the sequence of events leading to churn (e.g. billing dispute → service call → cancellation).
- No reason-for-churn data: The dataset records whether a customer churned but not why. Exit survey data or cancellation-reason codes would significantly improve targeting precision.
- Geographic coverage unknown: The dataset does not specify the provider's service region, limiting the ability to contextualise geographic churn clusters against competitive activity or network quality maps.
- No competitor data: Churn driven by competitor acquisition promotions cannot be isolated from churn driven by dissatisfaction. This limits the precision of the pricing and service recommendations.
- Static model: The Random Forest model was trained on a historical snapshot. Churn behaviour evolves; without a scheduled re-training pipeline, model performance will degrade over time.

Additional Data Recommended

Customer Support Interactions	Call/chat logs, CSAT scores, number of complaints, and resolution times — likely the strongest short-term churn signal not currently captured.
Network Quality Metrics	Outage frequency, downtime per customer, and speed-test performance, particularly for fiber subscribers.
Competitive Pricing Data	Local competitor pricing at time of churn decision, to separate value-driven vs. satisfaction-driven churn.
Exit Survey Responses	Cancellation reason codes from churned customers (price / service quality / moving / competitor offer / other).
Product Usage Data	Data consumption, call minutes, streaming hours — low usage often precedes churn and would improve model recall further.
Payment History	Late payments, failed payments, and payment method changes as leading churn indicators.

7. Next Steps

Immediate (0–30 days)

Deploy the Random Forest model scores to the CRM via the Tableau Operational dashboard. Train the retention team on the risk-score queue workflow. Launch the Contract Migration Campaign targeting month-to-month customers with tenure < 12 months.

Short-term (30–90 days)

Implement the First-90-Day Onboarding Programme. Launch the Fiber Optic Value NPS survey. Establish a weekly model refresh pipeline with automated score exports to the CRM.

Medium-term (90–180 days)

Integrate customer support interaction data into the feature set and retrain the model. Begin A/B testing of retention intervention scripts (discount offer vs. service upgrade). Measure and report retention campaign ROI against the \$722K baseline target.

Long-term (180+ days)

Build a longitudinal churn-event sequence model (e.g. survival analysis or LSTM) once time-series transaction data is available. Expand the Tableau suite to include A/B test tracking and retention campaign attribution. Develop an automated model re-training trigger when recall drops below 78% on live validation data.

New Model to Consider

A survival analysis model (Cox Proportional Hazards) would complement the current classifier by estimating not just who will churn, but when—enabling time-to-churn prioritisation in the outreach queue. This requires longitudinal event data currently unavailable but recommended for collection (see Section 6).